

Eike S. Eberhard

PhD Candidate in Machine Learning for Electronic Structure
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Machine-learned force fields simulate complex chemistry at a fraction of the cost of quantum mechanics, but they are fundamentally bounded by their training data. That data is the bottleneck: gold-standard coupled cluster cannot scale past small molecules, while standard DFT trades away accuracy. I attack this bottleneck by rethinking DFT itself. I train neural exchange-correlation functionals end-to-end through the self-consistent field solver, unlocking hybrid-level accuracy where hybrids are computationally out of reach. Then I make these calculations fast. Learned initializations slash the iterations needed to converge, while direct density-matrix prediction bypasses the solver loop entirely. My goal is to make coupled-cluster-quality data cheap enough to generate at the scale modern force fields demand.

KEY PUBLICATIONS

* equal contribution

- 2026 Oct **Physics-Aligned Electronic Ground-State Learning Improves Generalization**
E. S. Eberhard*, X. Kainz*, V. Kotsev, A. Aldossary, S. Günnemann
Preprint 2026 · ROCKET, GROOT, ON-Loss
- 2026 Jun **Derivative Informed Learning of Exchange-Correlation Functionals**
E. S. Eberhard, L. A. Thiede, A. Aldossary, A. Burger, N. Gao, V. Bhethanabotla, A. Aspuru-Guzik, S. Günnemann
ICML 2026 · DI-XC, DI-Loss
- 2026 Apr **Transferable SCF-Acceleration through Solver-Aligned Initialization Learning**
E. S. Eberhard*, V. Kotsev*, T. Güthle, S. Günnemann
NeurIPS 2026 · SAIL, ERIC
- 2026 Jan **3D Molecule Generation from Rigid Motifs via SE(3) Flows**
R. Poletukhin, M. Kollovich, **E. S. Eberhard**, S. Günnemann
NeurIPS 2026 · MotiFlow
- 2025 Apr **Force Generation by Enhanced Diffusion in Enzyme-Loaded Vesicles**
E. S. Eberhard, L. Burger, C. L. Pastrana, H. Seyed-Allaei, G. Giunta, U. Gerland
Nano Letters
- 2024 Oct **Learning Equivariant Non-Local Electron Density Functionals**
E. S. Eberhard*, N. Gao*, S. Günnemann
ICLR 2025 (Spotlight) · EG-XC

EDUCATION

GPA on the German scale from 1.0 (best) to 4.0 (passed).

- 2025 Jan – present **PhD Candidate**
Data Analytics and Machine Learning Group, TUM
Munich Data Science Institute (MDSI)
- 2026 Jun – Sep **Research Stay**
University of Toronto & Vector Institute
ML exchange-correlation functionals at the MatterLab (Prof. Aspuru-Guzik)
- 2022 Oct – 2024 Aug **M. Sc. Computer Science & Engineering** GPA 1.1
Technical University of Munich (TUM)
Best of class 2024
Thesis: *Machine Learning for Density Functional Theory*, Data Analytics and Machine Learning Chair (Prof. Günnemann)
- 2021 Oct – 2023 Oct **M. Sc. Physics** GPA 1.2
Technical University of Munich (TUM)
Specialization in Theoretical Biophysics
Thesis: *Effects of Enhanced Diffusion in Enzyme-loaded Vesicles: A Multiscale Simulation Study*
- 2017 Oct – 2021 Oct **B. Sc. Physics** GPA 1.8
Technical University of Munich (TUM)
Thesis: *Extending Dispersive Bounds to Include Sub-threshold Branch Cuts* (Mathematical Particle Physics)

WORK EXPERIENCE

- 2023 Nov – 2024 Feb **Chair of Theory of Complex Biosystem**
Writing a paper on the results of my master's thesis and integrating my simulation code into JAX-MD
- 2021 Sep – 2022 Jun **Deloitte Consulting GmbH**
Industrial software development standards, and machine learning software for business applications (*trufa*)
- 2020 Aug – 2021 May **Chair of Theoretical Elementary Particle Physics**
Research in theoretical particle and high-energy physics
Contributor to [EOS Software](#)
- 2019 Jun – 2020 Jan **Walter Schottky Institute**
Fundamental research on semiconductors
Automated reproducible data analysis and independently developed software correlating multiple sources of experimental data

INVITED TALKS

- 2026 Nov **Grimme Group**
University of Bonn
- 2026 Nov **Learning-Based Methods in Scientific Computing**
RICAM, Linz
- 2025 Nov **AMLab**
Ensing & Welling groups, University of Amsterdam
- 2025 May **University of Heidelberg**
- 2024 Nov **Learning on Graphs Reading Group**
MIT (online)

AWARDS

- 2025 Nov **ChemAI Best Poster Award**
Amsterdam, Netherlands
- 2016 Jun **Abitur Award**
German Physical Society